

Wave 4.2 Quantile Probabilistic Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the approved `wave 4.2` dispersion-aware campaign: quantile and Gaussian probabilistic regression for locally dispersed TE curves.

The campaign completed all planned entries

- `6` completed runs;
- `0` failed runs;
- `2` probabilistic profiles;
- `3` required direction surfaces: `global`, `Fw`, and `Bw`.

The runner-level scalar first entry is `te_track2h_quantile_p10_p50_p90_bw`. The program-level scalar winner did not change: `te_periodic_gru_sequence_remote_Bw` remains the best registry entry.

Campaign Artifacts

Artifact	Path
Campaign output	<code>output/training_campaigns/2026-06-12-11-16-18_track2h_quantile_probabilistic_campaign_2026_06_12</code>
Leaderboard	<code>output/training_campaigns/2026-06-12-11-16-18_track2h_quantile_probabilistic_campaign_2026_06_12/campaign_leaderboard.yaml</code>
Best-run pointer	<code>output/training_campaigns/2026-06-12-11-16-18_track2h_quantile_probabilistic_campaign_2026_06_12/campaign_best_run.yaml</code>
Execution report	<code>output/training_campaigns/2026-06-12-11-16-18_track2h_quantile_probabilistic_campaign_2026_06_12/campaign_execution_report.md</code>
Planning report	<code>doc/reports/campaign_plans/track_2/2026-06-12-00-01-04_track2h_quantile_probabilistic_campaign_plan_report.md</code>
Campaign package	<code>config/training/track2h_quantile_probabilistic_modeling/campaigns/2026-06-12_track2h_quantile_probabilistic_campaign</code>

Execution Summary

Field	Value
Campaign name	<code>track2h_quantile_probabilistic_campaign_2026_06_12</code>
Completed runs	<code>6</code>
Failed runs	<code>0</code>
Model type	<code>curve_aware_harmonic_residual_offset_probe</code>
Profiles	<code>quantile_p10_p50_p90</code> , <code>gaussian_n11</code>
Tested surfaces	<code>global</code> , <code>Fw</code> , <code>Bw</code>

Field	Value
Runner-level scalar first entry	te_track2h_quantile_p10_p50_p90_bw
Program-level scalar winner changed	no

Directional Branch Results

Surface	Candidate	Profile	Curve	Test MAE	Test RMSE	Val MAE
global	gaussian_nll_global	gaussian_nll	mu	0.003013	0.003388	0.003267
Fw	gaussian_nll_fw	gaussian_nll	mu	0.003165	0.003548	0.003293
Bw	quantile_p10_p50_p90_bw	quantile	p50	0.002927	0.003519	0.003436

These rows are the branch-level closeout result. The Bw quantile row is also the campaign scalar leader, but it does not replace the global or Fw branch.

Probabilistic Leaderboard

Rank	Surface	Candidate	Profile	Test MAE	Test RMSE	Val MAE
1	Bw	quantile_p10_p50_p90_bw	quantile	0.002927	0.003519	0.003436
2	Bw	gaussian_nll_bw	gaussian_nll	0.002998	0.003608	0.003298
3	global	gaussian_nll_global	gaussian_nll	0.003013	0.003388	0.003267
4	Fw	gaussian_nll_fw	gaussian_nll	0.003165	0.003548	0.003293
5	Fw	quantile_p10_p50_p90_fw	quantile	0.003285	0.003668	0.003269
6	global	quantile_p10_p50_p90_global	quantile	0.003383	0.003764	0.003606

Calibration Snapshot

Surface	Candidate	Test Coverage	Test Width	Extra Diagnostic
global	quantile_p10_p50_p90_global	0.849	0.011449	crossing 0.000
Fw	quantile_p10_p50_p90_fw	0.849	0.010536	crossing 0.000
Bw	quantile_p10_p50_p90_bw	0.710	0.007536	crossing 0.000
global	gaussian_nll_global	0.747	0.008761	sigma 0.003418
Fw	gaussian_nll_fw	0.903	0.011687	sigma 0.004560
Bw	gaussian_nll_bw	0.758	0.008887	sigma 0.003467

The quantile heads did not exhibit quantile crossing in the serialized metrics. Coverage is direction-dependent: Fw is well covered for both families, while Bw achieves the best deterministic MAE with narrower and lower-coverage intervals.

Robust-Loss Comparison

Surface	Probabilistic Best	Prob. MAE	Robust Best	Robust MAE	Delta
global	gaussian_nll_global	0.003013	mae_robust_global	0.003406	+11.53%
Fw	gaussian_nll_fw	0.003165	mae_robust_fw	0.003146	-0.60%
Bw	quantile_p10_p50_p90_bw	0.002927	smooth_l1_robust_bw	0.003074	+4.77%

The probabilistic package is therefore stronger than the robust-loss package on scalar training metrics for `global` and `Bw`. It is not stronger for `Fw`, where the robust `mae` candidate remains slightly ahead.

Profile Interpretation

Profile	Observed Signal	Interpretation
quantile_p10_p50_p90	Best campaign scalar result on <code>Bw</code> ; no quantile crossing.	Promising for backward dispersion, but needs TE Curve Verification Pipeline curve playback before promotion.
gaussian_nll	Best <code>global</code> and <code>Fw</code> probabilistic branch results.	Useful as a stable uncertainty-aware control and possibly the better <code>global</code> / <code>Fw</code> probabilistic default.

The main modeling result is not that probabilistic heads solve TE Curve Verification Pipeline. The useful result is that explicit uncertainty-aware heads can beat the first robust-loss package on `global` and `Bw` scalar metrics, while exposing calibration diagnostics that robust point losses cannot provide.

Registry Effects

The campaign runner refreshed family registries for all six probabilistic families and updated the program registry. The program-level scalar best did not move: `te_periodic_gru_sequence_remote_Bw` remains the current scalar program winner with test `MAE` = 0.002344.

Registry Scope	BestRun	Test MAE
Wave 4.2 global probabilistic	te_track2h_gaussian_nll_global	0.003013
Wave 4.2 Fw probabilistic	te_track2h_gaussian_nll_fw	0.003165
Wave 4.2 Bw probabilistic	te_track2h_quantile_p10_p50_p90_bw	0.002927

TE Curve Verification Pipeline Boundary

Official TE curve-first verification was not run as part of this normal campaign closeout. Under campaign governance, that remains a separate operator-approved workflow after this campaign-results report and PDF are complete.

The next TE Curve Verification Pipeline package should add all six probabilistic candidates and report accepted results separately for:

- `global` ;
- `Fw` ;
- `Bw` .

The verification must compare raw curve error, centered-shape error, offset, amplitude, harmonic behavior, collage plots, and overlay plots against Wave 3.3, Wave 4.1 robust-loss, Wave 3.2, Wave 2.2, and the current accepted TE Curve Verification Pipeline baselines.

Closeout Decision

Wave 4.2 quantile/probabilistic execution is complete: all planned candidates have successful training artifacts, no failed run remains, registries were refreshed by the runner, and the active campaign state can be cleared.

From a modeling standpoint

- carry `gaussian_nll_global` forward as the strongest probabilistic global scalar candidate;
- carry `gaussian_nll_fw` forward as the strongest probabilistic forward scalar candidate, while noting it is slightly behind robust `mae_robust_fw` ;
- carry `quantile_p10_p50_p90_bw` forward as the strongest probabilistic backward scalar candidate;
- keep both quantile and Gaussian heads alive until official TE Curve Verification Pipeline curve verification decides whether scalar gains translate into curve quality;
- do not promote Wave 4.2 probabilistic candidates before official TE Curve Verification Pipeline verification.

Recommended Follow-Up

1. Accept this closeout and clear the active campaign state.
2. Prepare a separate operator-launched TE curve verification refresh for all six probabilistic candidates.
3. If the official TE Curve Verification refresh confirms the scalar `global` and `Bw` gains, carry probabilistic heads into later mixture-density, latent-state, and multi-head design.
4. Keep robust `mae_robust_fw` visible as the current stronger `Fw` dispersion-aware scalar candidate.
5. Keep `Fw` , `Bw` , and `global` as parallel branch decisions.